

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **predictive analytics pipelines** using **Python (pandas, scikit-learn, NumPy)** and **SQL** to forecast payment transaction volumes across enterprise banking systems, reducing operational forecasting error by **18%** and enabling proactive capacity planning for **500M+ daily SWIFT messages**.
- Built **Power BI** and **Tableau** dashboards consolidating real-time monitoring data from **Splunk, CloudWatch, and Dynatrace** across **Java Spring Boot** microservices and **AWS infrastructure (EC2, S3, Lambda)**, delivering **SLA compliance visibility** to C-suite stakeholders and reducing incident detection time by **35%**.
- Developed **machine learning classification models** using **scikit-learn** and **statistical hypothesis testing** to identify anomalous payment patterns in **ISO 20022 message flows** and **SEPA cross-border transactions**, improving fraud-detection accuracy to **94%** precision while maintaining **GDPR** compliance and regulatory audit trails.
- Engineered **ETL data pipelines** in **Python** and **SQL** (PostgreSQL, Oracle, MySQL) integrating **ERP systems** with enterprise business platforms, automating reconciliation workflows and reducing manual data-validation effort by **22 hours weekly** across banking operations teams.
- Led **statistical analysis** and **A/B testing** using **R/RStudio** to optimize payment-processing workflows and user-experience enhancements, delivering **8 quantified business-impact reports** that drove prioritization of **15+ feature releases** in Agile sprints managed via **Jira**.
- Implemented **CI/CD automation** with **Jenkins** and **Git** for reproducible model deployment; version-controlled analytics assets in **Confluence**; and established **data governance** standards across cross-functional teams (engineers, business analysts, compliance), enabling **24-hour** model-to-production cycles.

SKILLS

Data Science & Advanced Analytics – Machine learning (supervised/unsupervised), statistical hypothesis testing, A/B testing, predictive modelling, regression analysis, classification, time-series forecasting, scikit-learn, evaluation metrics, feature engineering.

Data Engineering & Pipeline Development – ETL design and implementation, data warehousing, SQL optimization, Python (pandas, NumPy), data validation, schema design, PostgreSQL, MySQL, Oracle, data governance.

Business Intelligence & Visualization – Power BI (dashboards, DAX, Power Query, data modelling), Tableau, interactive reporting, KPI dashboards, stakeholder communication, data storytelling.

Statistical Methods & Languages – R/RStudio, exploratory data analysis, regression, hypothesis testing, probability distributions, A/B testing design, reproducible research.

Enterprise Software & Integration – Java Spring Boot, REST APIs, AWS (EC2, S3, Lambda, CloudWatch), Docker, microservices, ERP integration, system interconnectivity, change management.

Production Monitoring & Operations – Splunk, Dynatrace, CloudWatch, SLA management, incident response, application support, Linux, shell scripting, log analysis, alerting strategies.

Development & Collaboration Tools – Jenkins, Git, Jira, Confluence, Agile/Scrum, code review, CI/CD pipelines, version control, cross-functional workflows.

EDUCATION AND TRAINING

PROJECTS

Real-Time Payment Anomaly Detection & Predictive Analytics Platform (HCL Technologies)

- Architected a **machine learning-driven anomaly detection system** processing **ISO 20022** and **SWIFT MT/MX message flows** using **Python (scikit-learn, pandas, NumPy)** to classify fraudulent and irregular transactions in real time, achieving **94% precision** and **91% recall** across **500M+ daily payment messages** in a banking production environment.
- Engineered **predictive analytics pipelines** with **SQL (PostgreSQL, Oracle)** and **Python** to forecast payment volume spikes, transaction latency, and cross-border settlement delays **7 days in advance**, reducing operational surprises by **40%** and enabling proactive auto-scaling of **AWS Lambda** and **EC2** infrastructure.
- Built end-to-end **Power BI dashboards** and **Tableau visualizations** integrating real-time data from **Splunk, Dynatrace, CloudWatch** and internal **ERP systems**, surfacing model predictions, fraud alerts, and **SEPA/PSD2 compliance metrics** to banking operations and risk teams; dashboards consumed by **120+ daily active users** across 8 business units.
- Implemented **CI/CD automation** with **Jenkins** and **Git** to deploy model updates and **DAX-powered reports** into production **weekly**; established **GDPR-compliant data governance** standards in **Confluence** and trained 12 cross-functional stakeholders on model interpretation, reducing time-to-insight by **18 days per quarter**.
- Validated model performance using **statistical hypothesis testing, A/B testing**, and **R/RStudio** analysis; published **6 peer-reviewed technical reports** documenting feature importance, fairness audits, and regulatory alignment with **ISO 20022** compliance requirements, informing **3 major product roadmap decisions**.

SWIFT Payments Operations Dashboard & Reconciliation Intelligence (HCL Technologies)

- Designed and deployed a **production-grade monitoring dashboard** using **Power BI** and **Splunk** to visualize transaction-flow health, settlement status, and **ISO 20022 message validation** across **SEPA** and cross-border payment streams, enabling **P1/P2 incident detection within 2 minutes** and reducing **SLA breach incidents by 31%**.
- Engineered **ETL data pipelines** in **Python** and **SQL** to reconcile **10M+ daily payment records** between banking systems and **ERP platforms**, automating reconciliation workflows that previously required **40 FTE hours weekly** of manual validation and reducing data-discrepancy resolution time from **48 hours to 4 hours**.
- Collaborated with banking-domain business analysts and compliance teams to embed **GDPR-compliant audit trails**, transaction-level lineage tracking, and regulatory reporting into dashboard architecture, achieving **100% data-governance compliance** across payment operations and enabling **zero failed regulatory audits** over **18 months**.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)